

SCOUTING-MODEL DEMONSTRATION

Data-Driven Recruitment

Turning a real squad situation into role-fit, league-adjusted shortlists — the Celtic FC January 2026 reinforcement case.

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What this demonstrates — and what it doesn't

1 Squad context: real

Celtic's actual 2025-26 situation, measured from dated match data up to the January 2026 window.

2 Candidates: model demo

Scored on a provided StatsBomb-grade dataset. The data season is labelled on every shortlist; older data is never shown as current.

3 The question

Can the model surface the right player TYPES for the need? This is “what the engine finds” — not “sign this player now.”

The approach: a five-step scouting pipeline



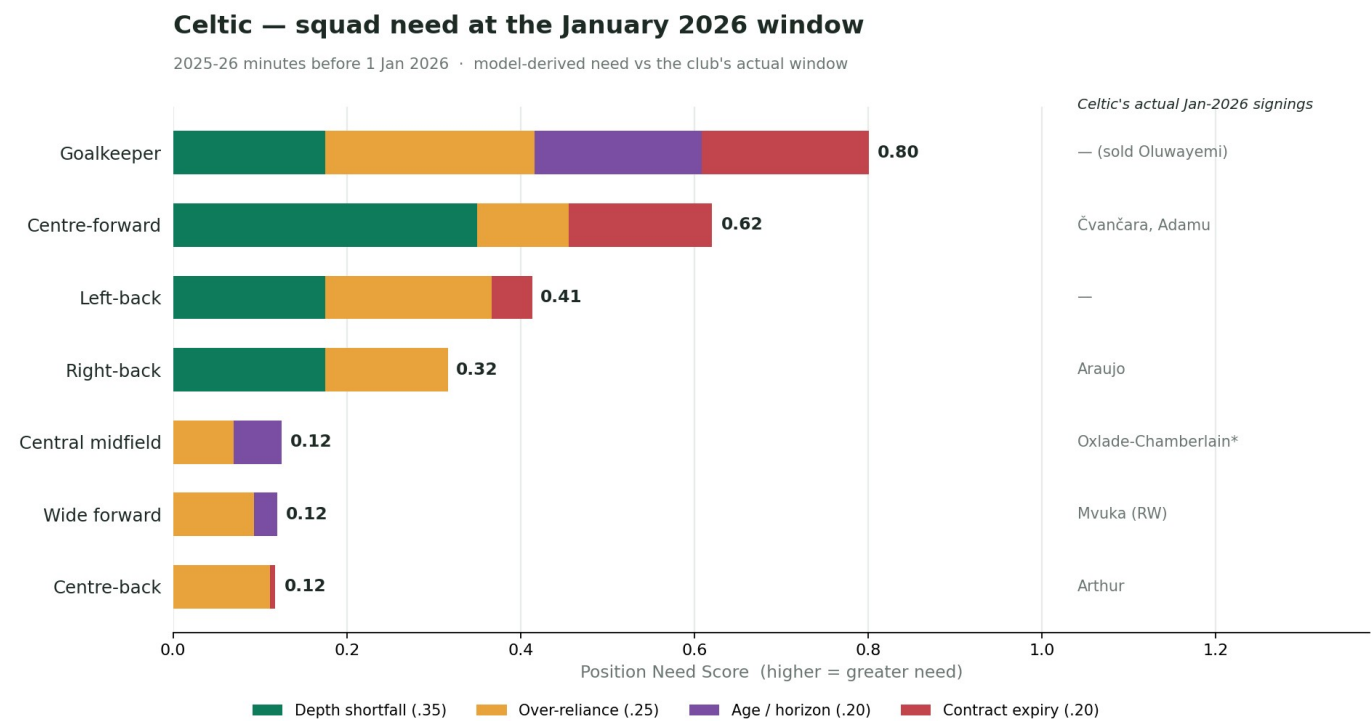
Each step is reproducible. Squad need is data-derived; roles are defined from the manager's system; scoring is league-adjusted; everything is split into permanent and loan tracks before validation.

Where the model fits — the PFSA scouting workflow



This pipeline operationalizes the six-stage PFSA Level 2 technical-scouting workflow: the model automates the data-heavy front end — need analysis, role-based KPI scoring and shortlisting — and produces the structured evidence the human stages act on.

Where Celtic needed to strengthen — and how it's measured



Need score = 4 weighted signals

- 0.35** **Depth shortfall** too few senior options / no rotation
- 0.25** **Over-reliance** minutes concentrated on one player = fragile if injured
- 0.20** **Age / horizon** ageing in the position, succession needed
- 0.20** **Contract** deals running down = loss risk

All four computed from dated minutes, ages and contracts — an objective structural signal.

From need score to priority — the logic

A Objective need score

The structural signal from the four weighted components. Tells you WHERE the squad is thin or fragile.

B Contextual overlay

Three judgements the raw score can't see: • profile diagnostics (foot / side gaps) • system importance under Nancy • January feasibility (loan vs buy, budget).

C Final priority

Need × context → the ranked action list the club should work to.

Resulting priority Goalkeeper (ageing) · Centre-forward (no rotation) · Full-backs · then centre-back & right wing by PROFILE, not raw volume.

The manager lens — Wilfried Nancy's system

The system

3-4-2-1 that becomes a 3-2-5 in possession.

Build from the back, deliberately invite the press, then break lines with quick combinations — and counter-press aggressively the moment the ball is lost.

What it demands of players

- Composure on the ball everywhere
- Ball-playing centre-backs who step into midfield
- Two-way wing-backs providing the width
- A mobile, pressing, link-and-create front line — his Columbus focal point was Cucho Hernández, not a static poacher

Role definitions across the squad

CF

Mobile link-9 (Cucho)

Drops, drifts, combines, presses, finishes

npXG · xGChain · xA · pressing · dribbling

WB

Engine of the width

Overlaps, delivers, carries, defends 1v1

xA · crosses · final-3rd carries · padj defending

CB

Ball-playing progressor

Builds from the back, steps in, progresses

xGBuildup · carries · pass% under pressure · padj defending

W

Interchanging creator

Combines, interchanges, creates, presses

xA · key passes · dribbles · npXG · pressing

GK

Ball-playing sweeper

Builds from the GK, sweeps, shot-stops

goals saved · pass% under pressure · sweep distance

Centre-forward role — built for Nancy, not a poacher

A poacher would over-index on box touches and finishing. Nancy's forward must also link, create, press and carry — so the weights change.

25%

Goal threat

Non-penalty xG per 90 — still scores, no longer dominant

20%

Link play

xGChain per 90 — involvement in build-up

20%

Creation

Expected assists per 90 — makes chances for others

20%

Pressing

Pressures per 90 — the counter-press trigger

15%

Carrying

Successful dribbles — drives and combines

Comparing players across leagues fairly

A goal or an xG in one league is not worth the same as in another. **Every output metric is scaled by a league-strength coefficient before players are compared.**

v1

Transparent coefficients

A documented prior per league (Premier League = 1.00), from public strength/market signals.

v2

Bridge-player method

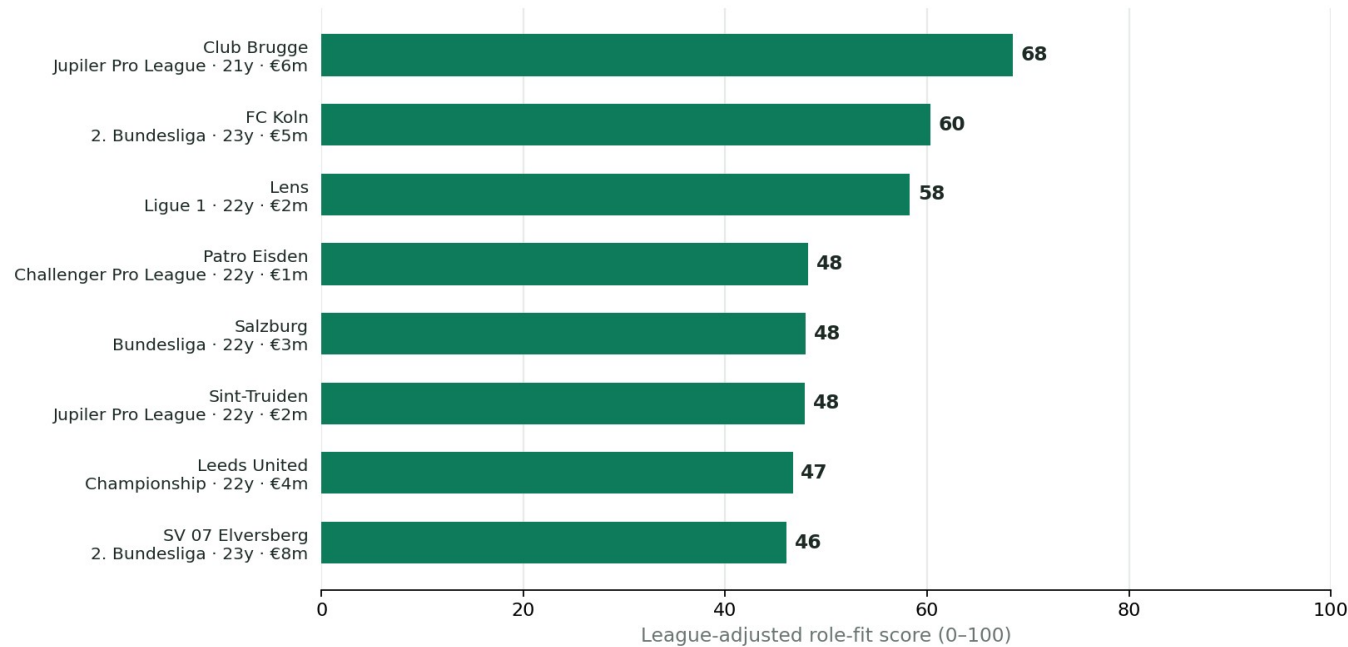
Estimate coefficients from how players' output changes when they move between leagues — the same direction as Hudl/StatsBomb player-impact models.

League	Coef.
Premier League	1.00
Bundesliga (GER)	0.92
Ligue 1	0.85
Championship	0.66
Jupiler Pro League	0.60
Premiership (SCO)	0.55
J1 League	0.50
Eliteserien	0.45
Allsvenskan	0.42

Track 1 — permanent: buy young, cheap, sell high

Permanent track — buy young, cheap, sell high

Centre-forward (Nancy/Cucho role) · age ≤ 23, value ≤ €8m · league-adjusted · data 2024-25



How it's built

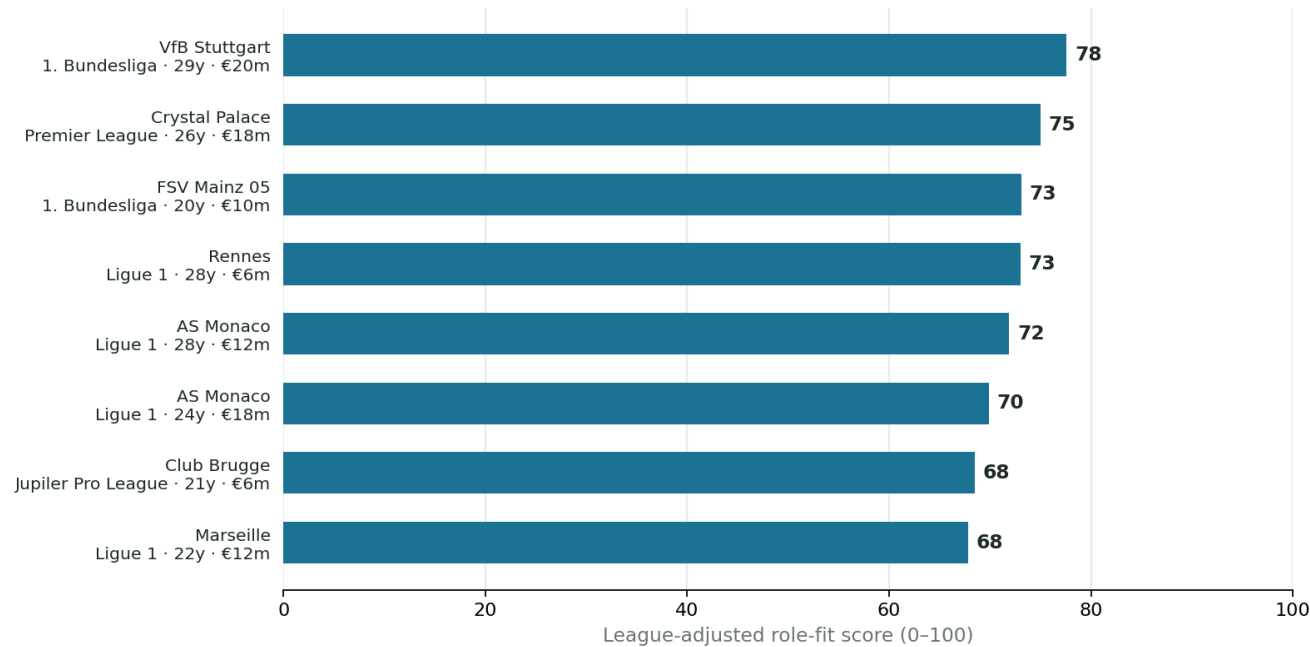
- Nancy/Cucho centre-forward role
- Age ≤ 23 and value ≤ €8m
- Re-weighting drops pure poachers; pressing, carrying, linking forwards rise (Patro Eisden, Sint-Truiden, Lens)
- Club Brugge's 21-y-o tops it — a complete profile

Data season: 2024-25 (labelled).

Track 2 — loan: ready-now impact, age-agnostic

Loan track — ready-now impact, age-agnostic

Centre-forward (Nancy/Cucho role) · value ≤ €20m · league-adjusted · data 2024-25



How it's built

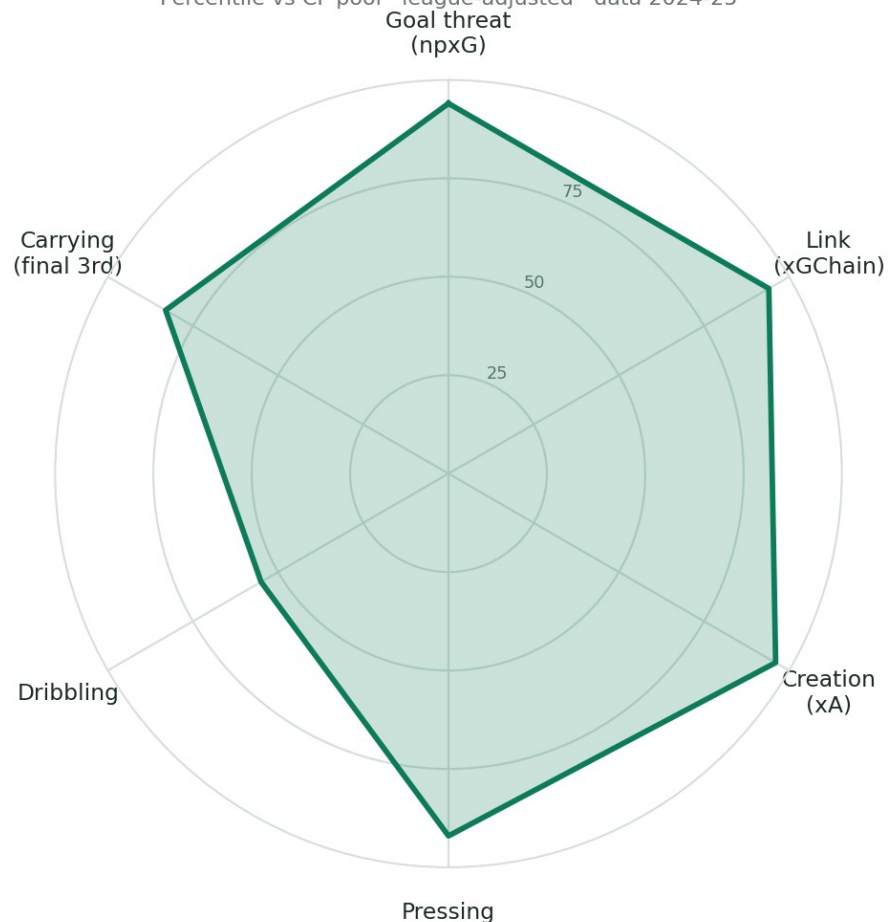
- Same role, age and contract ignored
- Targets quality squad players at stronger clubs
- Re-frames Celtic's real loan signings: judge underlying quality and fit, not minutes or age

Data season: 2024-25 (labelled).

Player profile — the buy-young pick

Top permanent pick — Club Brugge

Percentile vs CF pool · league-adjusted · data 2024-25

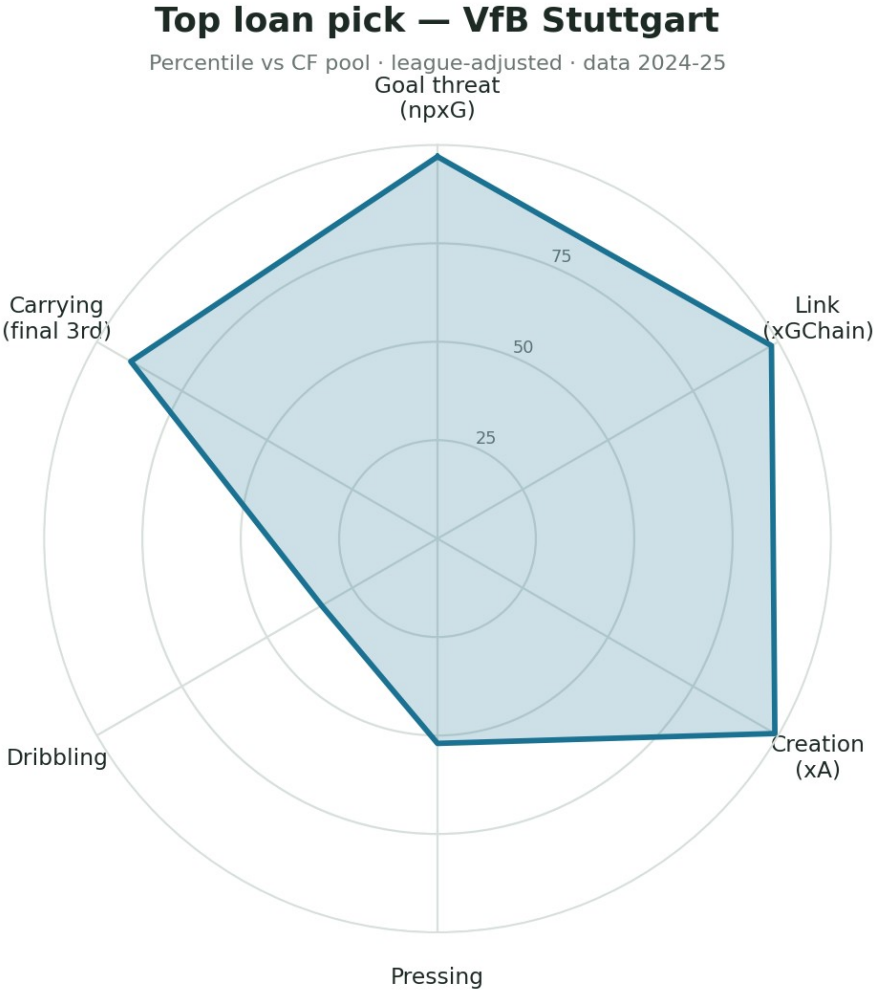


Club Brugge · 21 · €6m · Jupiler Pro League

A complete Nancy forward: elite goal threat (94th), link (94th) and creation (96th), with high-volume pressing (92nd) and final-third carrying (83rd). Combines, presses and finishes — the Cucho mould.

Archetype buy-young, sell-high: a complete profile in a strong feeder league.

Player profile — the ready-now pick



VfB Stuttgart · 29 · €20m · Bundesliga

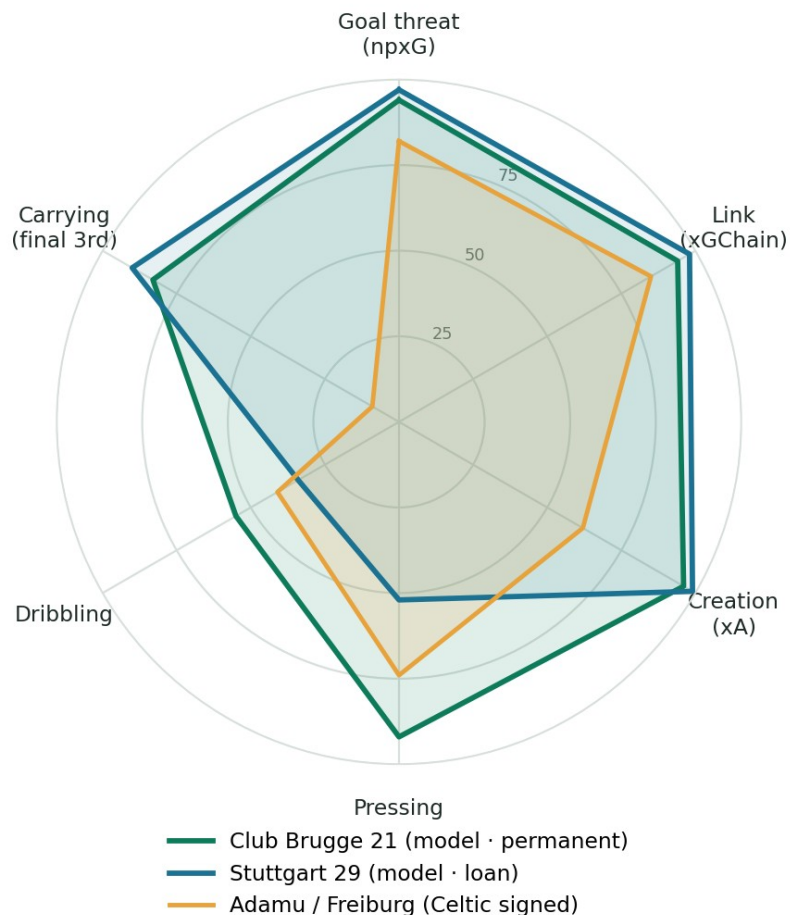
Elite threat (97th), link (98th), creation (99th) and final-third carrying (90th) — a deeply-involved creative No.9. Presses less than the Cucho ideal (52nd), so more orchestrator than harrier.

Archetype ready-now loan: proven Bundesliga output, fills the gap immediately.

Did the model match Celtic's actual window?

Model picks vs Celtic's actual CF signing

Nancy/Cucho centre-forward role · percentile vs CF pool · data 2024-25



Consistency verdict

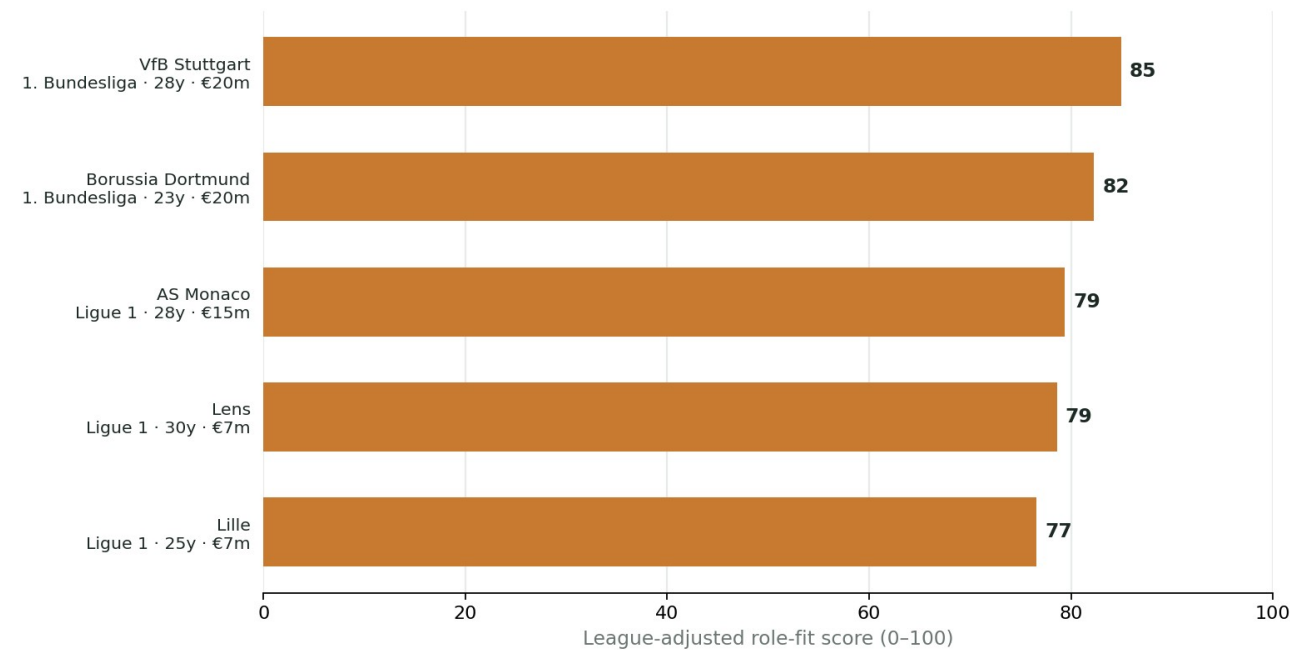
- ✓ **Need** — Model's top need was a centre-forward; Celtic signed two strikers.
- ✓ **Profile type** — Adamu fits the pressing (74th) and link (85th) parts of the Nancy forward — a real, if partial, fit.
- △ **Gaps** — He's weaker on carrying (9th) and creation — the Cucho hallmarks the model's picks max out.
- ✓ **Outcome** — Both loans delivered rotation minutes, addressing the depth gap the model flagged.

Actual H2 at Celtic Čvančara 778 min · 2G · 2A Adamu 230 min · 1G · 1A

Same engine — wing-back

Wing-back — engine of the width

External targets · two-way wing-back role · league-adjusted · data 2024-25



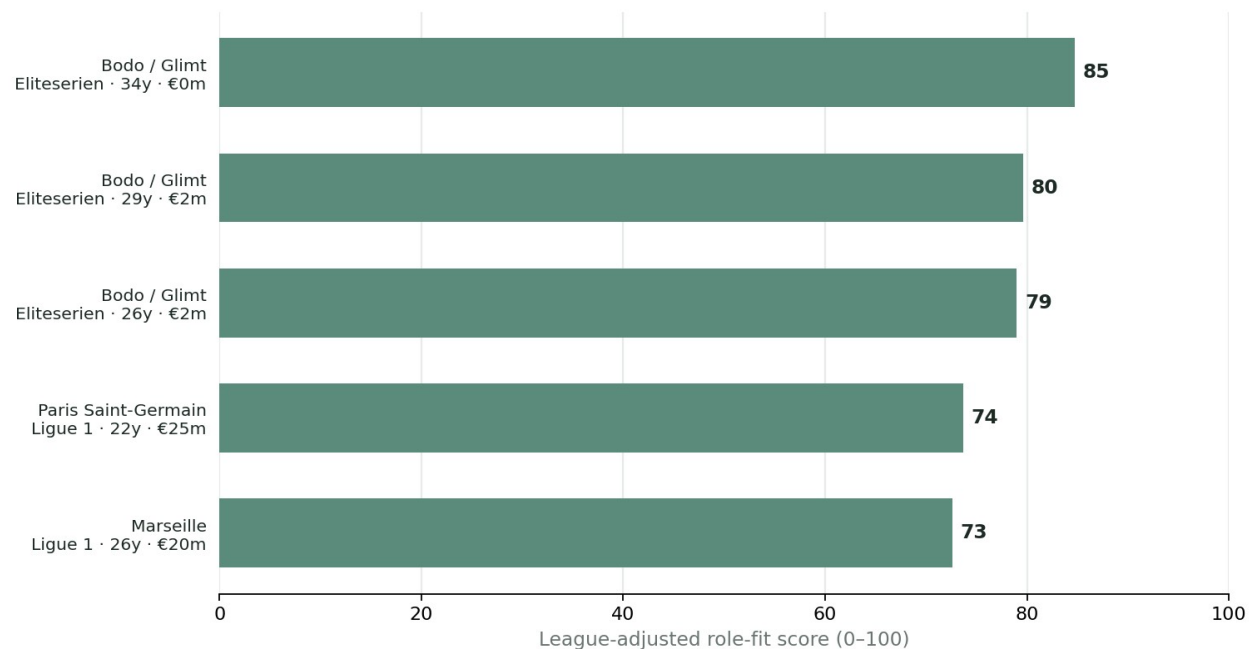
Engine of the width

- Two-way wing-back role — wide creation + progression + 1v1 defending
- Celtic's own full-backs score highly here, so it's less of a need — fittingly, Alistair Johnston was Nancy's wing-back at Columbus
- For the buy-young lens, Dortmund's 23-y-o leads the externals

Same engine — centre-back

Centre-back — ball-playing progressor

External targets · build-from-the-back role · league-adjusted · data 2024-25



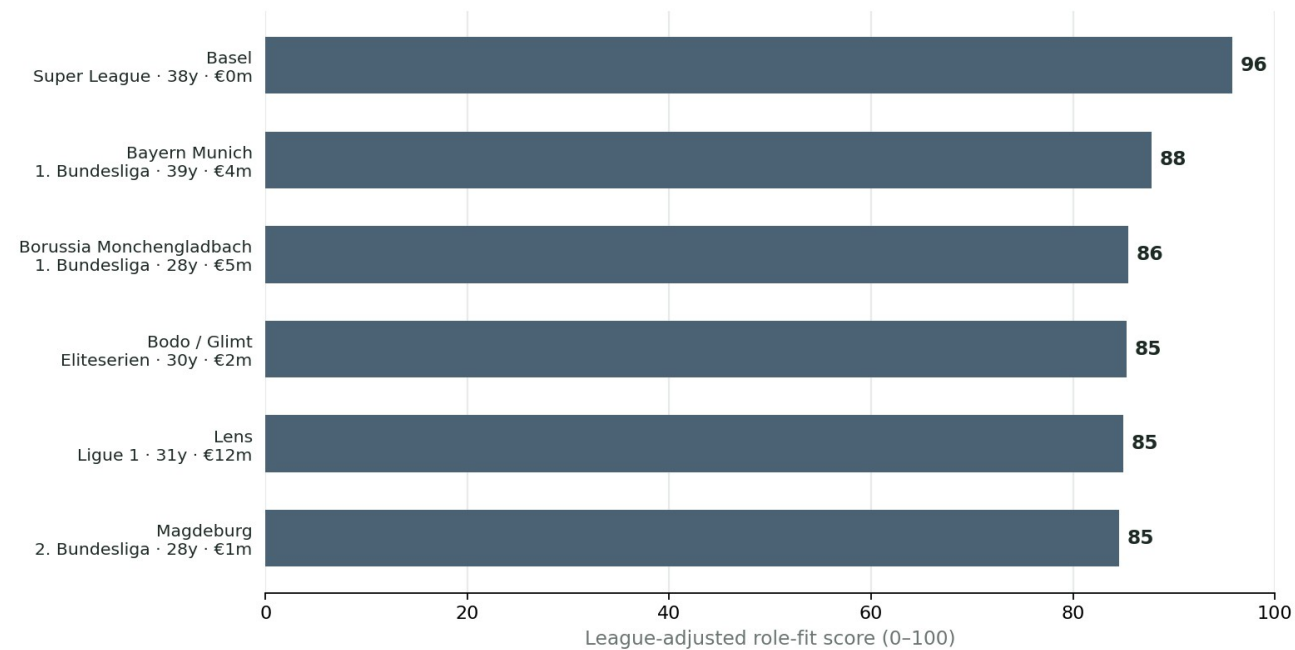
Ball-playing progressor

- Build-from-the-back role — progression + composure under press + defending
- Bodø/Glimt's possession centre-backs surface strongly — a Nancy-like system
- PSG's 22-y-o is the high-ceiling buy-young option; the real Celtic gap was a fit right-footed profile, not raw quality

Same engine — goalkeeper

Goalkeeper — ball-playing sweeper

External targets · distribution + sweep + shot-stopping · data 2024-25



Ball-playing sweeper

- Distribution under pressure + sweeping + shot-stopping
- The engine actually rates Celtic's current keeper top of this profile — but at 39, succession is the need
- Filtered for a young successor, Gladbach's and Magdeburg's 28-y-olds are the value targets

Why you can trust it — and where it stops

Rigour

- Roles defined from the manager's system, not templates
- League-strength adjustment across 15 leagues
- Transparent, adjustable KPI weights per role
- Reproducible code — every number re-derivable
- Human-in-the-loop: the model shortlists, people decide

Honest limits

- Candidate data is season-labelled, not live
- Players anonymised to IDs (team/league shown)
- Right-footedness & availability not in this dataset
- Output/xG ≠ full context — video check still required
- League coefficients & role weights are a first pass

Data & reproducibility

A Squad layer

davidcariboo Transfermarkt dataset — dated match appearances, real 2025-26 up to the January window. Powers the need analysis.

B Candidate layer

Provided StatsBomb-grade dataset — ~7,700 players, 15 leagues, full role KPIs (xG, xA, xGChain, pressures, GK metrics). Season labelled per shortlist.

Code & methods Python (pandas, scikit-learn) — need scoring, role-specific KPI models, league adjustment, two-track shortlists. Full implementation on GitHub.

A scouting pipeline, not a one-off list

It turns a real squad situation into manager-specific, league-adjusted, track-separated shortlists — transparent, reproducible, and built to run on a club's live data.